Interacting with me on behalf of REPLACE was Rajmohan, an argumentative and efficient engineer. His team worked tirelessly on the silica phenolic throat despite repeated failures—some of which led to my chiding Rajmohan who would argue back in full vigour, before promising to overcome the challenge and deliver the material.

After the Mahendragiri facility became operational—soon after the Vikas engine was successfully ground-tested at Vernon—we tested the throat by firing the engine several times. Most of the time, after half a minute, the throat would be ejected out of the nozzle. This became so frequent a failure that some at the test site would mock: 'It is forty seconds already. Why has the throat not come out?'

Rajmohan just would not give up. After having developed the material that can withstand the high temperature, when the throat was still not a success, the failure analysis team concluded that it was being pushed out because of its seating—a kind of groove or ridge on which the throat rested—which was giving way due to charring during the firing of the engine. In a few months, Rajmohan's team found the answer, and delivered the silica phenolic throat, which was used in the first flight of PSLV in 1993.

All this while, the Gowarikar-Muthunayagam war raged on. As the head of APSU, which worked on small control rockets and flow components, Muthunayagam did not have to report to Gowarikar, the VSSC director; but some of the liquid propulsion system work still came under VSSC. For this, too, Muthunayagam refused to see eye-to-eye with Gowarikar.

'You go ask him,' Muthunayagam would tell me whenever I approached him to take forward the Vikas project as part of the PSLV project.

After taking over as the VSSC director, Gowarikar had made goodwill visits to everyone, including Muthunayagam. This made me think that Gowarikar, now as the head of a prestigious institution, would have matured enough not to nurse a grudge against me. I was terribly wrong. Gowarikar opened an old chapter when I went to him with a request to retain the Vernon team so that we can develop the liquid propulsion engine without a hitch. The Vernon team, which was put together from different sections within VSSC, was by now back home and slowly disintegrating as they were called back to their previous assignments.